Speaker Information

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Talk Information

Mouse phenotype-driven approach to neuropsychiatric disorders

Behavioral analysis in mice allows evaluation of the effects of genetic and pharmacological manipulations at the organismal (whole-animal) level. In our laboratory, we investigate gene–brain–behavior relationships using a comprehensive behavioral test battery covering sensory, motor, emotional, and cognitive functions in various genetically modified mice. This approach is effective for screening models of neuropsychiatric disorders, and we have identified multiple lines with disease-relevant phenotypes. When similarities are observed not only in behavior but also in brain phenotypes, these models are more likely to reflect human pathology and may lead to the discovery of novel intermediate phenotypes. If such phenotypes are confirmed in patients, they can serve as new endophenotypes. This talk presents examples of phenotype identification based on behavioral analyses and optogenetic studies of underlying neural mechanisms, and discusses their implications for model development and therapeutic exploration.

Research Overview (for collaboration discussion)

Our laboratory aims to elucidate cellular and molecular mechanisms underlying cognitive and emotional processes, including memory and learning, using behavioral genetics, optogenetics, pharmacological and physiological approaches, and genome-edited mouse models. We investigate the pathophysiology of neuropsychiatric disorders and explore therapeutic strategies. In parallel, we develop reproductive technologies to support these efforts.

[Current Research Projects] Development of neuropsychiatric disorder models for pathophysiological analysis and therapeutic development; Functional elucidation of canonical and non-canonical genes; Physiological studies of meditation and Eastern practices

[Techniques] Mouse behavior analysis; Mouse reproductive engineering; Genome editing–based mouse generation; Optogenetics

[Aspects or techniques of interest for future collaboration]
We welcome collaborative projects involving the generation of genetically modified mice and mouse behavioral analysis.

Short Bio Prof. Keizo Takao graduated from The University of Tokyo (Dept. Psychology) and received a Ph.D. in Informatics (Kyoto University, 2006) and in Medical Sciences (University of Toyama, 2024). He has been a Professor at the University of Toyama since 2015 and currently serves as Chair of the Program of Cognitive and Emotional Neuroscience, and Director of the Division of Animal Resources and Development.